



GEV Hazard Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — GEV Hazard Model (MKM-GH-001)

Tests run on **23 March 2026 at 12:58:50**.

Table 1: Test Results Summary — GEV Hazard Model (MKM-GH-001)

Total Tests	16
Passed	16
Failed	0
Skipped	0
Duration	0.27s

Table 2: Detailed Test Results — GEV Hazard Model

Test Description	Acceptance Criteria	Result	Time
Exceedance probability in range	0.0 <= prob <= 1.0	Pass	0.000s
Fit returns three floats	isinstance(shape, float); isinstance(loc, float) (+1 more)	Pass	0.018s
Force gumbel sets shape zero	shape == 0.0	Pass	0.001s
Higher threshold lower probability	p_high < p_low	Pass	0.000s
Return level increases with period	r1_100 > r1_10	Pass	0.000s
Compute all responses returns dict	isinstance(responses, dict); len(responses) == len(synthetic_flat_gauges) (+1 more)	Pass	0.000s
Compute response returns gauge re- sponse	isinstance(response, GaugeResponse); response.gauge_id == gauge_id (+1 more)	Pass	0.000s
Response threshold flags	response.exceeded.alert; response.exceeded.warning	Pass	0.000s
Annual flood probabilities in range	0.0 <= curve.annual_flood_prob.alert <= 1.0; 0.0 <= curve.annual_flood_prob.warning <= 1.0 (+1 more)	Pass	0.082s
Build returns dict of curves	isinstance(curves, dict); len(curves) == len(synthetic_flat_gauges) (+2 more)	Pass	0.083s
Each curve is gauge hazard curve	isinstance(curve, GaugeHazardCurve); curve.gauge_id == gauge_id	Pass	0.090s
Cumulative prob increasing	probs[i] >= probs[i - 1]	Pass	0.000s
Probabilities sum to one	abs(point.survival_prob + point.prob.at.least.one - 1.0) ...	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Returns correct length	<code>len(ts) == 5</code>	Pass	0.000s
Returns term structure points	<code>isinstance(point, TermStructurePoint)</code>	Pass	0.000s
Survival prob decreasing	<code>survivals[i] <= survivals[i - 1]</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (16 tests)	D. Kelly
23-Mar-2026	1.1	16 test(s) added; 16 test(s) removed (total: 32)	D. Kelly